

# AI
# Term Project

Group 7
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## Outline

**01 Methodology**

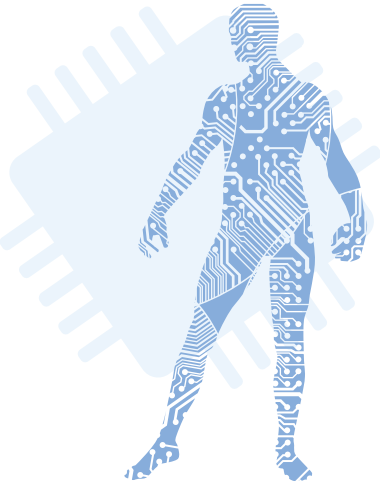
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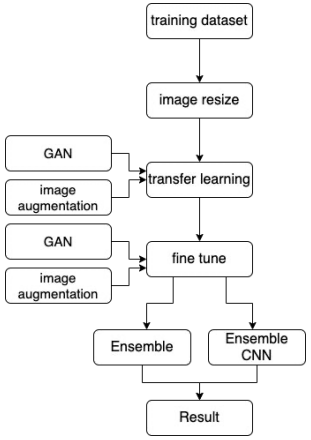
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## Methodology

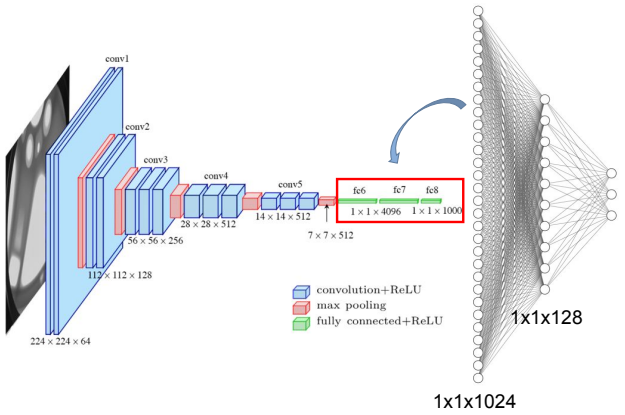
## Methodology




- Validation: 5% of training dataset
- Image resize: 224x224
- Transfer learning: multiple pretrained models
- Image augmentation
  - 15° rotation, 5% zoom range
  - CycleGAN
- Ensemble learning

# Evaluation & Test Result

## Transfer Learning




| Model | Accuracy |
|---|---|
| InceptionResNetV2 | 71.50% |
| VGG16 | 70.88% |
| Xception | 74.50% |
| ResNet101 | 54.87% |
| NASNetMobile | 71.50% |
| DenseNet121 | 74.25% |
| DenseNet169 | 74.12% |
| DenseNet201 | 74.50% |
| InceptionV3 | 70.12% |
| MobileNetV2 | 73.44% |

| Layer (type) | Output Shape | Param # |
|---|---|---|
| densenet121 (Model) | (None, 7, 7, 1024) | 7037504 |
| flatten (Flatten) | (None, 50176) | 0 |
| dense (Dense) | (None, 1024) | 51381248 |
| dropout (Dropout) | (None, 1024) | 0 |
| dense_1 (Dense) | (None, 128) | 131200 |
| dropout_1 (Dropout) | (None, 128) | 0 |
| dense_2 (Dense) | (None, 3) | 387 |

Total params: 58,550,339
Trainable params: 51,512,835
Non-trainable params: 7,037,504

## Hyper Parameter Tuning

| batch | 8 | 16 | 32 | 64 | 128 |
|---|---|---|---|---|---|
| epoch | acc | acc | acc | acc | acc |
| 30 | 72.75% | 74.25% | 71.00% | 74.12% | **75.12%** |
| 50 | 72.62% | 72.25% | 73.50% | 72.88% | 75.25% |
| 100 | 72.00% | 73.62% | 72.88% | 72.88% | 72.38% |
| 150 | 70.88% | 70.12% | 73.50% | 72.88% | 73.88% |
| 200 | 71.50% | 72.62% | 71.62% | 73.00% | 73.88% |

Learning Rate=**2.00E-05**

| batch | 8 | 16 | 32 | 64 | 128 |
|---|---|---|---|---|---|
| epoch | acc | acc | acc | acc | acc |
| 30 | 74.00% | 71.12% | 73.50% | 71.12% | 71.25% |
| 50 | 74.88% | 73.88% | 73.38% | 72.38% | 72.88% |
| 100 | 74.88% | 72.88% | 75.38% | 74.25% | 73.62% |
| 150 | 74.12% | 74.12% | 73.00% | 74.88% | 72.25% |
| 200 | 73.38% | 73.38% | 73.38% | 73.00% | 74.38% |

Learning Rate=2.00E-06

| batch | 8 | 16 | 32 | 64 | 128 |
|---|---|---|---|---|---|
| epoch | acc | acc | acc | acc | acc |
| 30 | 68.88% | 66.38% | 60.25% | 55.12% | 54.75% |
| 50 | 70.62% | 69.88% | 51.50% | 63.25% | 59.62% |
| 100 | 71.62% | 70.50% | 70.12% | 67.88% | 67.12% |
| 150 | 72.50% | 71.62% | 71.62% | 70.12% | 68.12% |
| 200 | 73.75% | 72.75% | 70.12% | 72.38% | 72.12% |

Learning Rate=2.00E-07

**Batch Size = 128**
**Epoch =30**
**Learning Rate=2.00E-05**

| Dropout | Accuracy |
|---|---|
| 0 | 0.7312 |
| 0.2 | 0.7425 |
| 0.3 | 0.7475 |
| **0.4** | **0.7525** |
| 0.5 | 0.7375 |
| 0.8 | 0.6825 |

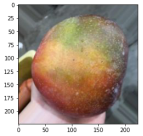
| batch | 8 | 16 | 32 | 64 | 128 |
|---|---|---|---|---|---|
| epoch | acc | acc | acc | acc | acc |
| 30 | 36.62% | 36.62% | 36.62% | 36.62% | 36.62% |
| 50 | 36.62% | 36.62% | 36.62% | 36.62% | 36.62% |
| 100 | 36.62% | 36.62% | 36.62% | 36.62% | 36.62% |
| 150 | 36.62% | 36.62% | 36.62% | 36.62% | 36.62% |
| 200 | 36.62% | 36.62% | 36.62% | 36.62% | 36.62% |

Learning Rate=2.00E-03

| batch | 8 | 16 | 32 | 64 | 128 |
|---|---|---|---|---|---|
| epoch | acc | acc | acc | acc | acc |
| 30 | 71.38% | 72.75% | 73.38% | 71.75% | 71.88% |
| 50 | 71.50% | 72.50% | 72.00% | 73.00% | 73.12% |
| 100 | 70.88% | 70.88% | 72.12% | 67.88% | 72.12% |
| 150 | 69.75% | 72.88% | 72.62% | 71.75% | 72.75% |
| 200 | 71.50% | 72.12% | 73.75% | 72.12% | 72.50% |

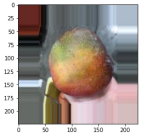
Learning Rate=2.00E-04

## Image Augmentation



Rotation Range

| **15** | 30 | 60 | 90 |
|---|---|---|---|
| **75.38%** | 74.50% | 72.75% | 71.00% |



Zoom Range

| **0.05** | 0.1 | 0.15 | 0.3 |
|---|---|---|---|
| **75.38%** | 74.50% | 72.75% | 71.00% |



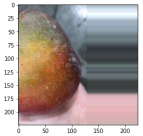
Horizontal Flip

| TRUE | FALSE |
|---|---|
| 72.88% | 75.38% |



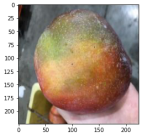
Vertical Flip

| TRUE | FALSE |
|---|---|
| 74.12% | 74.88% |



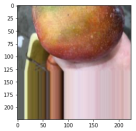
Width Shift Range

| 0.05 | 0.1 | 0.15 | 0.3 |
|---|---|---|---|
| 73.75% | 73.38% | 73.38% | 72.62% |



Shear Range

| 0.05 | 0.1 | 0.15 | 0.3 |
|---|---|---|---|
| 74.62% | 74.50% | 74.88% | 74.75% |

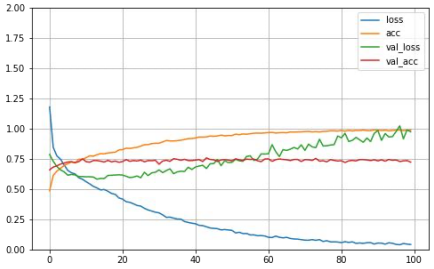


Height Shift Range

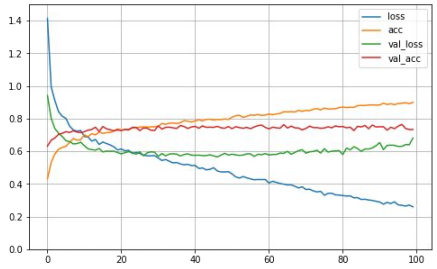
| 0.05 | 0.1 | 0.15 | 0.3 |
|---|---|---|---|
| 74.25% | 73.50% | 73.38% | 72.75% |

# Image Augmentation

Overfitting



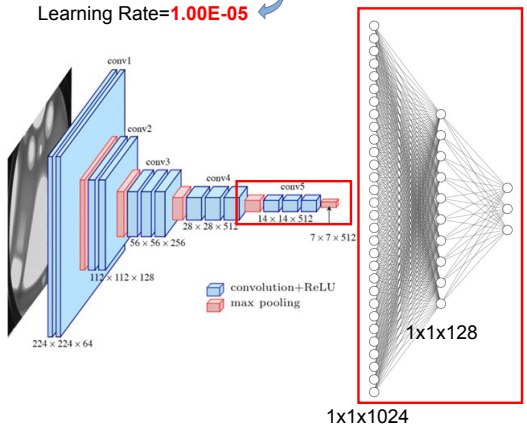

After augmentation




# Fine Tuning

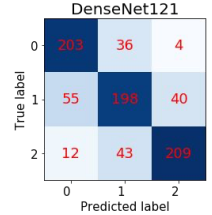
Learning Rate=2.00E-05

Learning Rate=1.00E-05

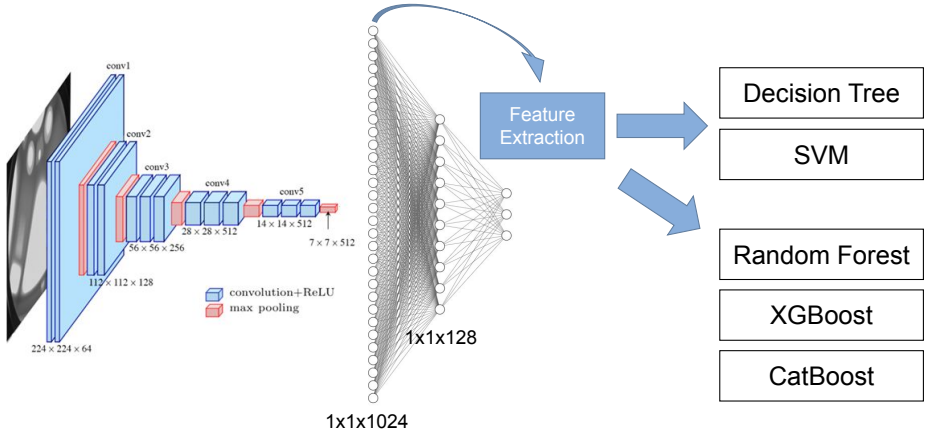



| | Before | After |
|---|---|---|
| DenseNet121 | 76% | 76.62% |
| Xception | 76.25% | 75.62% |
| MobileNetV2 | 73.44% | 74.87% |
| VGG16 | 74.62% | 75.88% |
| InceptionResNetV2 | 72.25% | 71.75% |
| NASNet | 72.25% | 73.38% |

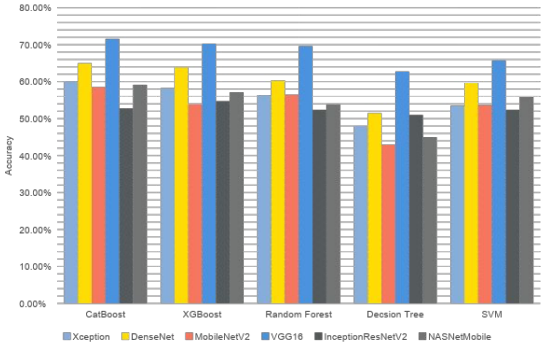
DenseNet121




# Ensemble Learning




# Ensemble Learning



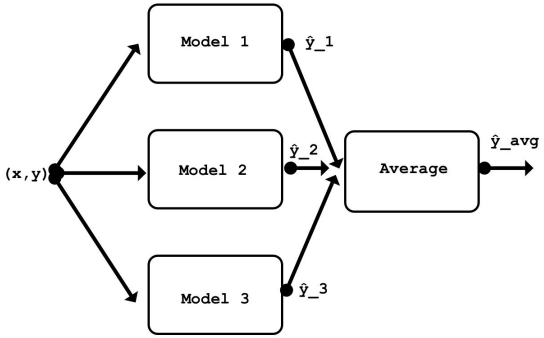

Accuracy

| | CatBoost | XGBoost | Random Forest | Decsion Tree | SVM |
|---|---|---|---|---|---|
| Xception | 60.13% | 58.25% | 56.38% | 48.00% | 53.50% |
| DenseNet | 65.00% | 63.88% | 60.38% | 51.50% | 59.63% |
| MobileNetV2 | 58.50% | 53.88% | 56.50% | 43.00% | 53.63% |
| VGG16 | 71.63% | 70.25% | 69.63% | 62.75% | 65.75% |
| InceptionResNetV2 | 52.75% | 54.75% | 52.38% | 51.00% | 52.38% |
| NASNetMobile | 59.13% | 57.13% | 53.88% | 45.00% | 55.88% |

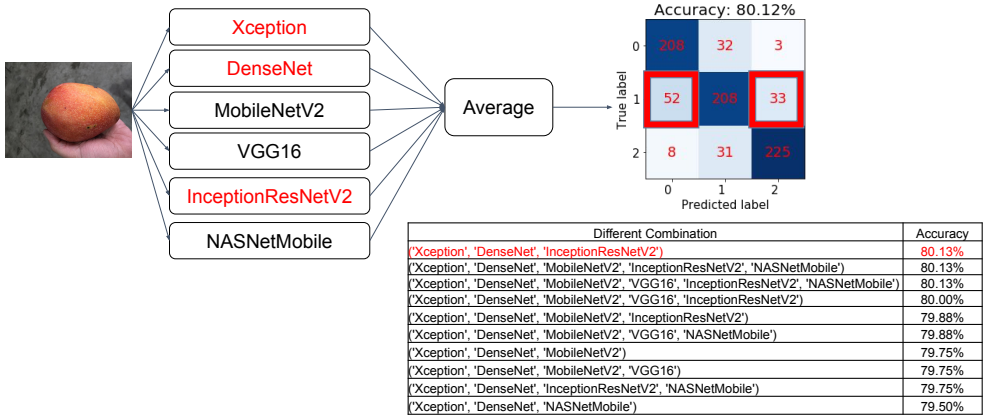
Grid Search & Cross validation

| | CatBoost | XGBoost | Random Forest | Decsion Tree | SVM |
|---|---|---|---|---|---|
| Xception | lr:0.1,md:6 | eta:0.1,md:6 | entropy,md:10,n:900 | entropy,md:6 | c:0.3 |
| DenseNet | lr:0.25,md:6 | eta:0.2,md:6 | entropy,md:10,n:900 | gini,md:6 | c:0.5 |
| MobileNetV2 | lr:0.1,md:6 | eta:0.1,md:6 | gini,md:10,n:900 | gini,md:6 | c:0.4 |
| VGG16 | lr:0.1,md:6 | eta:0.15,md:6 | entropy,md:10,n:500 | gini,md:6 | c:0.2 |
| InceptionResNetV2 | lr:0.05,md:6 | eta:0.25,md:6 | entropy,md:10,n:900 | entropy,md:6 | c:0.4 |
| NASNetMobile | lr:0.15,md:7 | eta:0.15,md:6 | entropy,md:10,n:900 | gini,md:6 | c:0.6 |

# Ensembling ConvNets

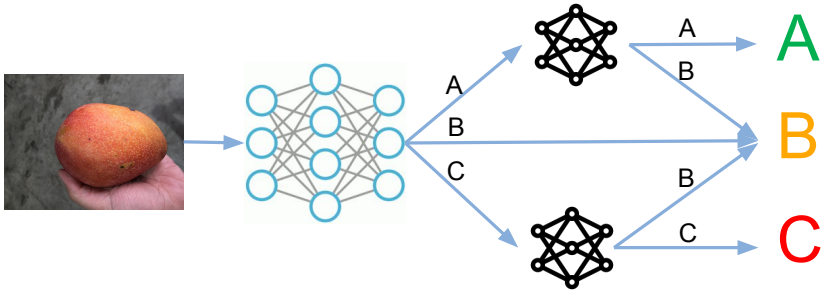



# Ensembling ConvNets

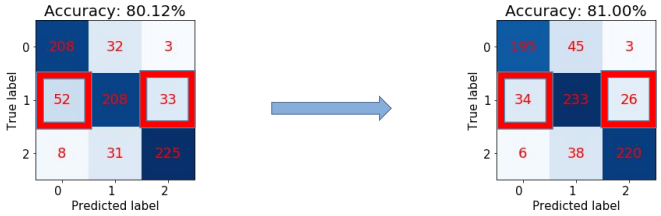



| Different Combination | Accuracy |
|---|---|
| ('Xception', 'DenseNet', 'InceptionResNetV2') | 80.13% |
| ('Xception', 'DenseNet', 'MobileNetV2', 'InceptionResNetV2', 'NASNetMobile') | 80.13% |
| ('Xception', 'DenseNet', 'MobileNetV2', 'VGG16', 'InceptionResNetV2', 'NASNetMobile') | 80.13% |
| ('Xception', 'DenseNet', 'MobileNetV2', 'VGG16', 'InceptionResNetV2') | 80.00% |
| ('Xception', 'DenseNet', 'MobileNetV2', 'InceptionResNetV2') | 79.88% |
| ('Xception', 'DenseNet', 'MobileNetV2', 'VGG16', 'NASNetMobile') | 79.88% |
| ('Xception', 'DenseNet', 'MobileNetV2') | 79.75% |
| ('Xception', 'DenseNet', 'MobileNetV2', 'VGG16') | 79.75% |
| ('Xception', 'DenseNet', 'InceptionResNetV2', 'NASNetMobile') | 79.75% |
| ('Xception', 'DenseNet', 'NASNetMobile') | 79.50% |

# AB + BC cascaded classifier



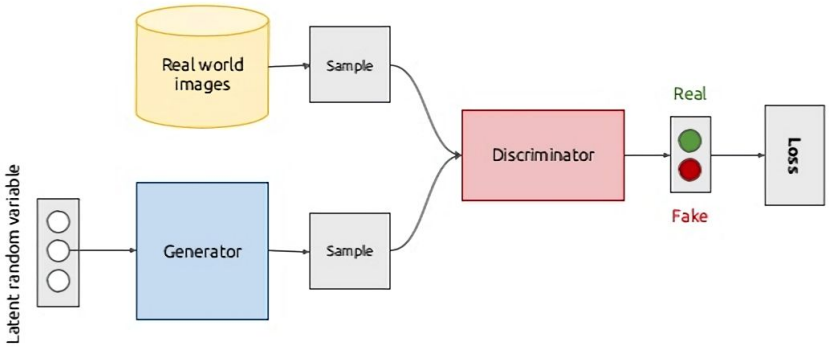

# AB + BC cascaded classifier




| A, B classifier | |
|---|---|
| Different combinations | Accuracy |
| ('Xception', 'DenseNet', 'MobileNetV2') | 0.833955224 |
| ('Xception', 'DenseNet', 'NASNetMobile') | 0.832089552 |
| ('Xception', 'MobileNetV2', 'VGG16', 'InceptionResNetV2') | 0.830223881 |
| ('Xception', 'DenseNet', 'MobileNetV2', 'VGG16', 'InceptionResNetV2') | 0.830223881 |
| ('Xception', 'DenseNet', 'MobileNetV2', 'VGG16', 'NASNetMobile') | 0.828358209 |

| B, C classifier | |
|---|---|
| Different combinations | Accuracy |
| ('Xception', 'MobileNetV2', 'InceptionResNetV2') | 0.867145 |
| ('Xception', 'DenseNet', 'MobileNetV2', 'InceptionResNetV2') | 0.863555 |
| ('MobileNetV2', 'InceptionResNetV2', 'NASNetMobile') | 0.861759 |
| ('MobileNetV2', 'VGG16', 'InceptionResNetV2', 'NASNetMobile') | 0.861759 |
| ('Xception', 'DenseNet', 'MobileNetV2') | 0.859964 |

# GAN data augmentation

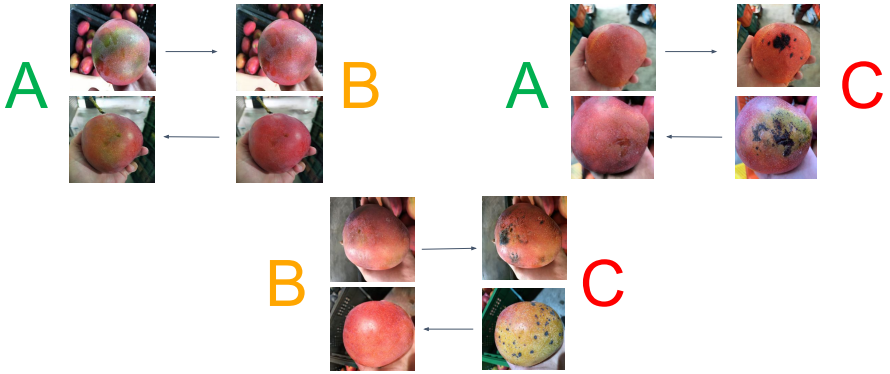



# DCGAN





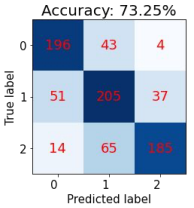
# CycleGAN

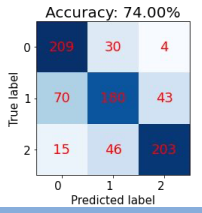



# CycleGAN (Class B → Class C)



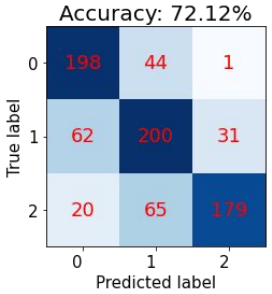




## CycleGAN (Class B → Class A)






## CycleGAN (Class C → Class A)





## CycleGAN (Class A → Class C)

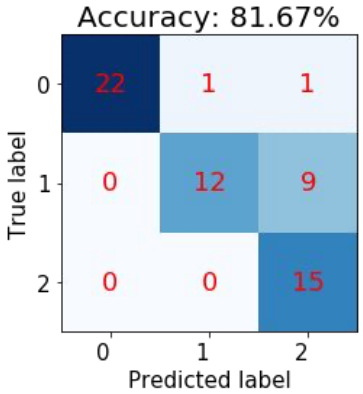




## Demo Result

# Demo Result

- About the same accuracy as test result




# Discussion

# Discussion

- Ensemble of CNNs performs better than CNN feature extraction + traditional ML methods
  - using dimensionality reduction methods to select best features might improve the result
- GAN
  - DCGAN: architecture may be improved
  - CycleGAN: generated images might be the “easy” case

# Conclusion

# Conclusion

- Best performing model:
  - Some image augmentation
  - Ensemble of Xception, DenseNet, InceptionResNetV2
  - Cascaded with AB and BC classifiers
- Aggregate Different model can improve the result
- Learning rate will extremely affect the accuracy



Thanks for your attention!